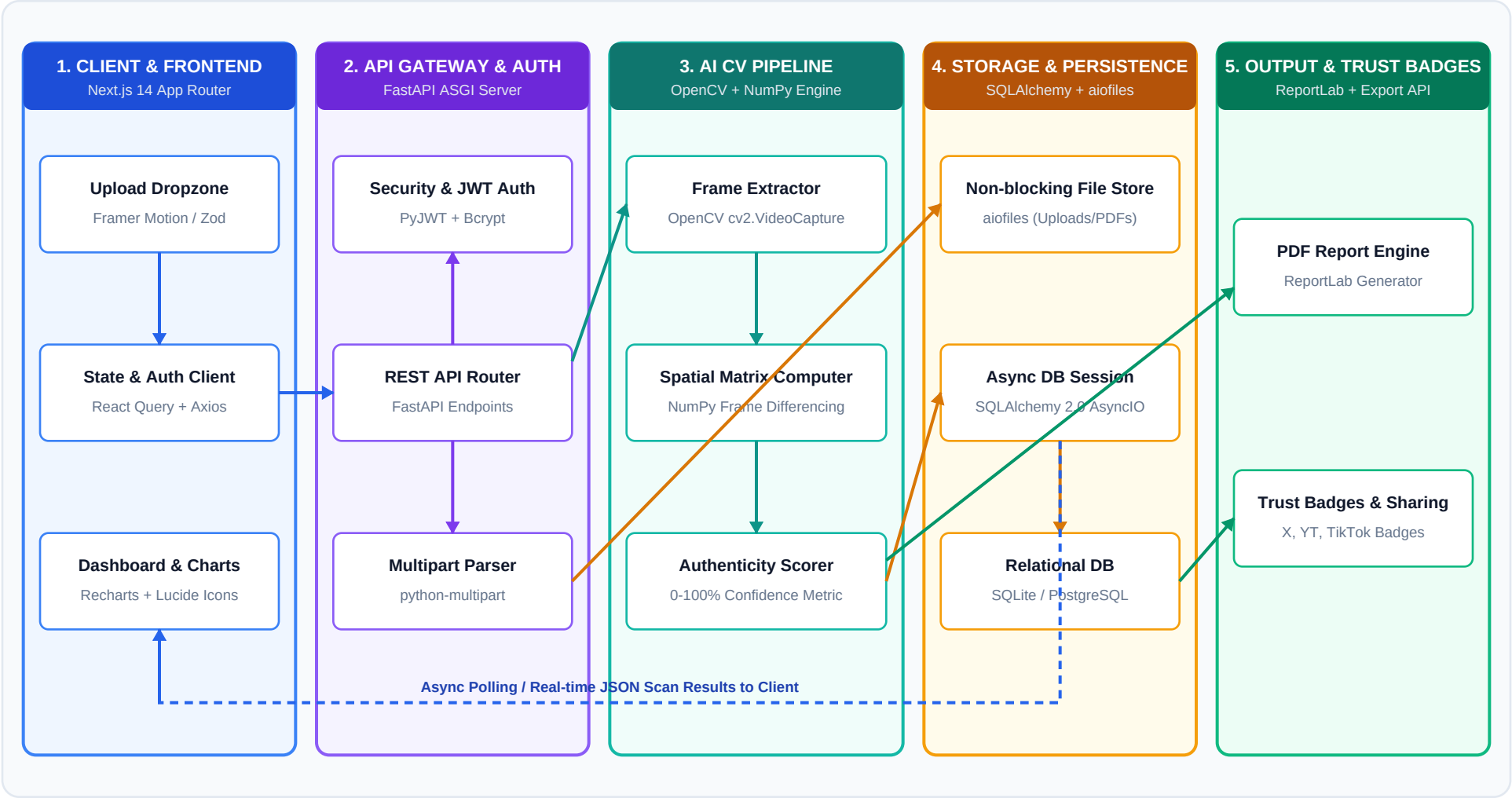


TruthLens ■■ — System Architecture & End-to-End Data Pipeline

Visual Architectural Diagram, Layer Interactions & Execution Flowchart



System Architecture Component & Flow Reference Matrix

This reference matrix breaks down the key stages illustrated in the architecture flowchart, detailing the exact responsibility, inputs, outputs, and protocol used at each step.

Pipeline Stage	Component & Subsystems	Protocol / Communication	Input -> Output Artifacts
Stage 1: Client & UI	Next.js 14, Framer Motion, TanStack Query, Recharts	HTTPS / REST JSON API calls via Axios	Raw Video File (.mp4/.mov) -> Upload Progress & Real-time Charts
Stage 2: API Gateway	FastAPI, PyJWT, Bcrypt, python-multipart	ASGI Asynchronous Worker Loops	HTTP Multipart Stream -> Validated Session & File Handle
Stage 3: AI CV Pipeline	OpenCV (Headless), NumPy Array Math, Pillow	In-memory NumPy Matrix Transformation	Video Frames -> Spatial Noise & 0-100% Authenticity Score
Stage 4: Persistence	SQLAlchemy 2.0 AsyncIO, aiofiles, SQLite/Postgres	Async DB Connection Pool (`aiosqlite` / `asyncpg`)	Scan Payload -> Immutable DB Rows & Server Storage
Stage 5: Output Badges	ReportLab PDF Generator, Social Trust Badges API	PDF Document Stream & Public Verification Links	Analysis Record -> Downloadable Audit PDF & Social Badges

Key Architectural Highlights & Performance Optimizations

- 1. Fully Asynchronous Event Loop:** Fast API + `uvicorn` coupled with `SQLAlchemy AsyncIO` ensures zero blocking thread delays while reading videos or running database queries.
- 2. Headless Computer Vision Processing:** `OpenCV-headless` avoids unnecessary GUI dependencies on servers, allowing fast memory-efficient frame slicing and NumPy matrix transforms.
- 3. Client-Side Optimistic UI:** Next.js 14 with `TanStack React Query` handles polling and loading states gracefully, rendering real-time Recharts animations as soon as scan scores are ready.
- 4. Cryptographic Auditability:** Every scan outcome generates a cryptographic signature and an exportable PDF report, ensuring anti-tamper verification for enterprise audits.